

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

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In 2022–23 the Netherlands raised its countercyclical capital buffer (CCyB) to 2% while the rulebook said 0%. It was refilling a buffer, not fighting a boom

SUMMARY

De Nederlandsche Bank (DNB) raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023, binding since 31 May 2024). At both decisions the Basel reference indicator, the credit-to-GDP gap, stood far below zero (−33 and −47 percentage points), so the rulebook pointed to a buffer of 0%.

Q1 · ENVIRONMENT

The economy had recovered from COVID, but war, an energy shock and 11.6% inflation darkened the outlook, and the ECB began raising rates in July 2022. Credit was cold, housing was hot and then cooled, and the banks were strong and growing more profitable.

Q2 · CRITERIA AND RISKS

DNB's 2022 framework sets 2% as the normal level, calibrated on past crisis losses rather than on credit growth. The buffer insures against shocks that no indicator predicts, while housing risks are handled by other tools. It also kept a promise made in March 2020.

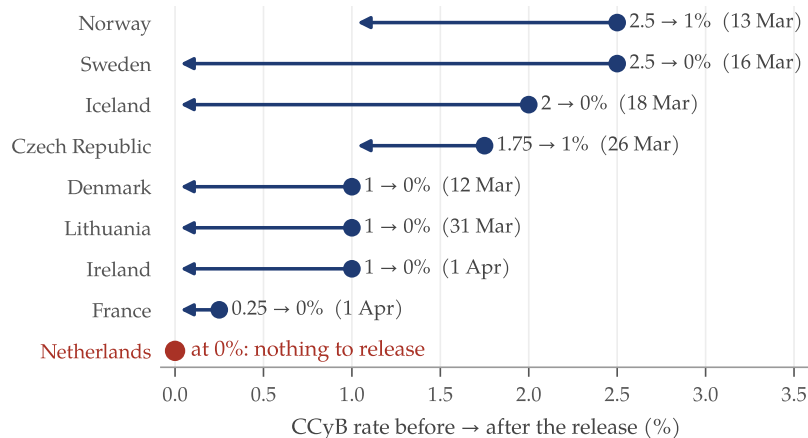
Q3 · IMPLICATIONS

The buffer was cheap, at most about a fifth of the big banks' spare capital. It was largely a swap for cuts in structural buffers, and we see no sign of a contraction in lending. The Netherlands now has €6.7bn it can release.

Our verdict: **normalisation, not tightening**. The price is a greater reliance on DNB's judgement, and the buffer has not yet been tested in a crisis.

In March 2020 COVID struck and neighbours released their buffers, but the Dutch buffer was empty

PROLOGUE



Within three weeks, eight European countries cut their countercyclical buffers so that banks could absorb losses and keep lending. Belgium and Germany cancelled increases they had already announced.

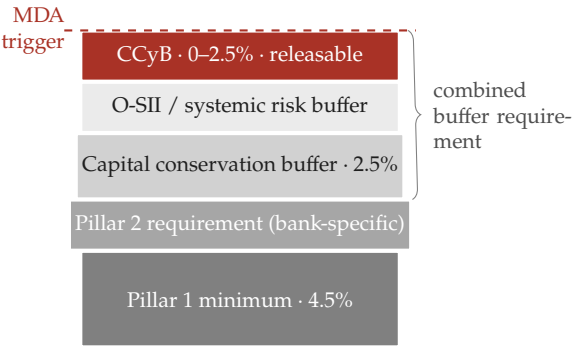
The Netherlands had nothing to release. DNB improvised: on 17 March 2020 it cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. This freed **€8bn** of capital, enough to support up to **€200bn** of lending.

“Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%.”

DNB, 17 March 2020

Source: ESRB, CCyB rates table (announcements March–April 2020); DNB press release “DNB lowers bank buffer requirements to support lending”, 17 Mar 2020.

That mattered because the CCyB is the only layer of bank capital an authority can release in a crisis



Capital stack in % of risk-weighted assets (RWA), not to scale.

Read the stack from the bottom. Banks must always meet the minimum requirements; the buffers on top can be used in bad times.

If a bank's capital falls below the top of its buffers (the MDA trigger), its dividends and bonuses are restricted automatically. Banks therefore avoid using buffers even when they are allowed to. In COVID, banks close to this line cut lending instead.

The CCyB works differently. The authority can switch it off, so the requirement itself falls and the bank faces no penalty and no stigma. Increases bind only after twelve months, but a release is immediate.

Only a buffer that the authority can **release** is truly usable in a crisis, and it has to be filled beforehand.

Two years later DNB refilled it in the middle of a storm and against its own rulebook. Was that the wrong moment?

THE QUESTION

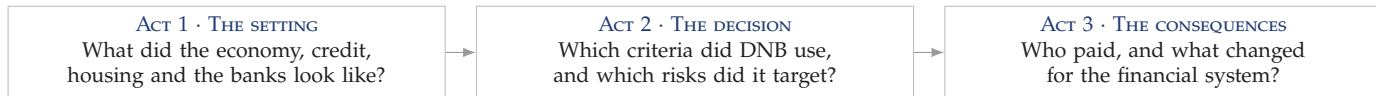
THE STORM OF 2022–23

Russia invaded Ukraine and energy prices hit records. Dutch inflation averaged 11.6% in 2022, the ECB raised rates from July 2022, and house prices were about to fall. The Basel rule, based on the credit gap, said the buffer should be 0%.

THE DECISION

DNB nevertheless raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023). Each step became binding a year later, in May 2023 and May 2024.

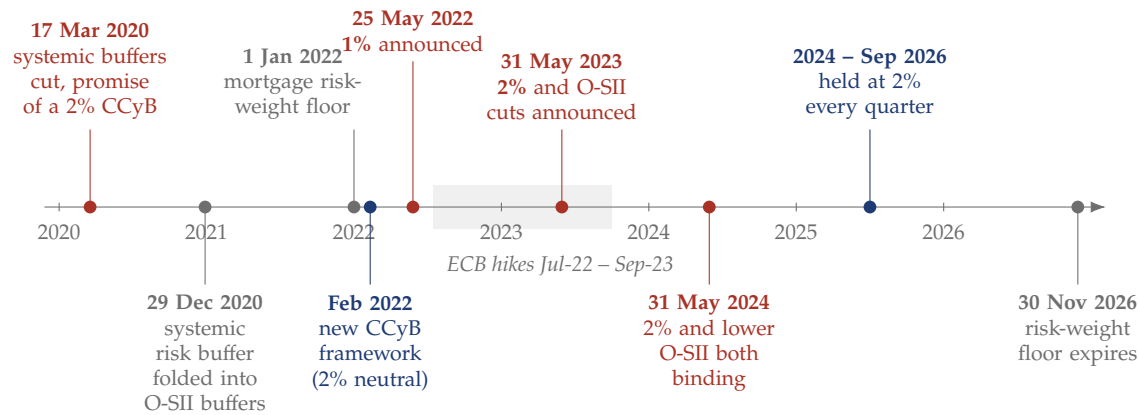
Raising capital requirements in a downturn can amplify the cycle, which is exactly the **procyclical mistake** the CCyB was invented to avoid. So was DNB tightening into a slowdown, fighting a hidden boom, or doing something else?



Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).

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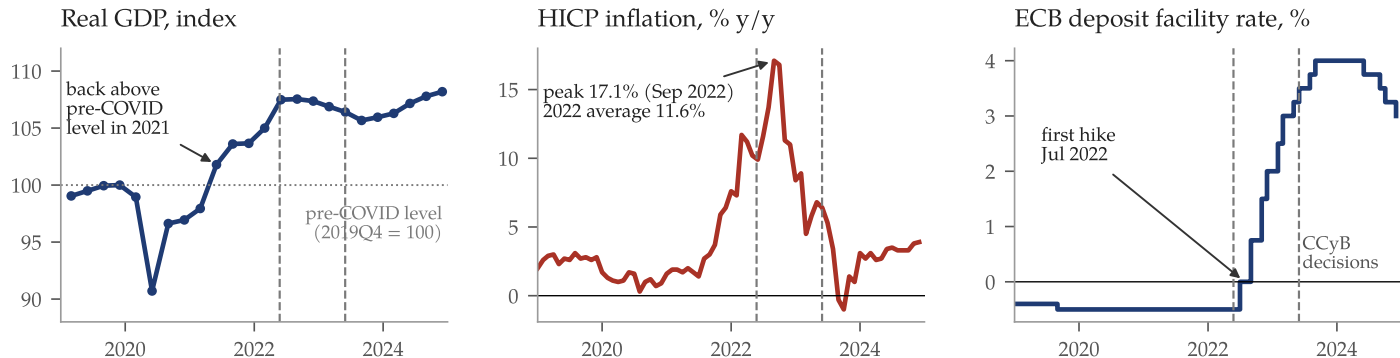
The refill followed the path promised in 2020, one step per year, just as the ECB started to tighten



DNB moved in two equal steps of one percentage point, a year apart, and then stopped at 2%. The first step was announced two months before the ECB's first rate hike.

By then the economy had fully recovered, but war, energy prices and inflation were darkening the outlook

Q1 · ENVIRONMENT



GDP was back above its pre-COVID level in 2021, which is the condition DNB's framework sets before a buffer is built. Then inflation rose to a peak of 17.1%, and the ECB moved its deposit rate from -0.5% to 4% in fourteen months. DNB wrote in spring 2022 that *"the economic outlook has worsened due to the war in Ukraine and high inflation."*

The macro picture argued both ways: build the buffer while the economy can take it, or wait because it is weakening.

Source: ECB Data Portal (Dutch real GDP, MNA; Dutch HICP, ICP; deposit facility rate, FM), current vintage; DNB Financial Stability Report, spring 2022. Dashed lines: CCyB announcements.

In the financial system credit was cold and only housing was hot, and even housing soon turned

Q1 · ENVIRONMENT



Housing was hot. Nominal prices rose 19% in the year to early 2022, and DNB itself wrote that *“the overheated housing market calls for further measures.”* As mortgage rates rose, real prices fell by 9%.

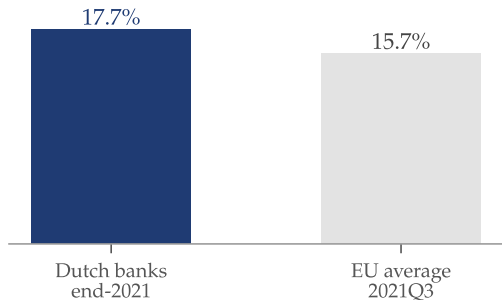
Credit was not. Household debt is high (107% of GDP against 58% in the euro area) but it was falling, and the share of income spent on debt service was the lowest since 2005.

If there was a boom, it was in house prices, not in credit.

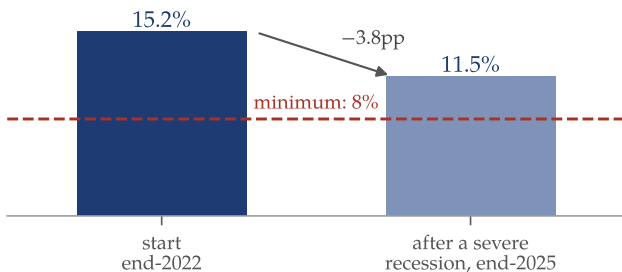
Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Dashed lines: CCyB announcements (May 2022, May 2023).

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Core capital (CET1) ratio, % of RWA



DNB stress test, four largest banks' CET1 ratio

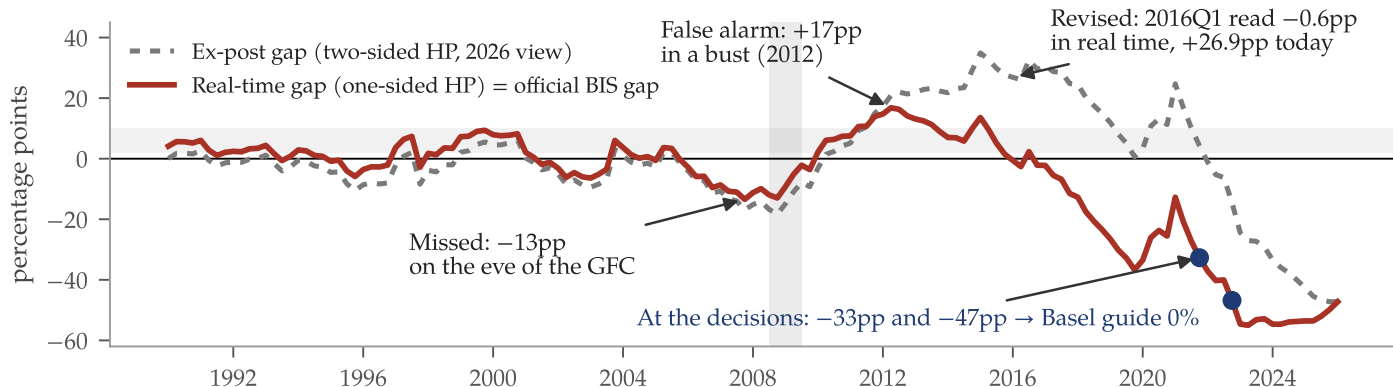


Dutch banks held more core capital than the EU average, and in DNB's stress test the four largest banks stayed well above the minimum after a severe recession. Rising rates also lifted their net interest income, by 6.7% in 2022; DNB had expected higher rates to *"ultimately have a positive impact on the profitability of financial institutions."*

Act 1 in one sentence: war, inflation, rate hikes and a turning housing market argued against raising the buffer, while a recovered economy, cold credit and strong banks argued for it.

So why ignore the rulebook? On Dutch data the Basel rule had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS



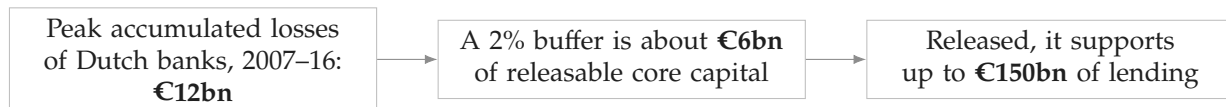
The credit-to-GDP gap measures how far credit relative to GDP is above its long-run trend. Under the Basel guide the buffer starts when the gap exceeds 2 points and reaches 2.5% at 10 points.

On Dutch data the gap missed the 2008 crisis, sounded a false alarm in 2012 and was later revised. At the two decisions it read -33 and -47 points.

Source: BIS credit-to-GDP gaps (Q.NL.PA); Drehmann et al. (2010); Edge & Meisenzahl (2011); own one-sided HP filter ($\lambda = 400,000$).

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DNB used its own compass instead: 2% in normal times, calibrated on past crisis losses rather than on credit growth



The framework sets the buffer at 2% when risks are neither high nor low, a so-called *positive neutral* buffer. DNB reads a dashboard of indicators but stresses that there is “*no mechanical link . . . guided discretion.*” Seventeen jurisdictions now work this way, and Sweden, the UK and Poland also target 2%.

The 2% is sized to absorb about half of past crisis losses. It does not depend on credit growth.

Its stated reasons were recovery, uncertainty and the 2020 promise, never credit growth

Q2 · CRITERIA & RISKS

MAY 2022 · FROM 0% TO 1%

*“When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB.”*

“The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up.”

DNB judged cyclical risks to be normal to elevated.

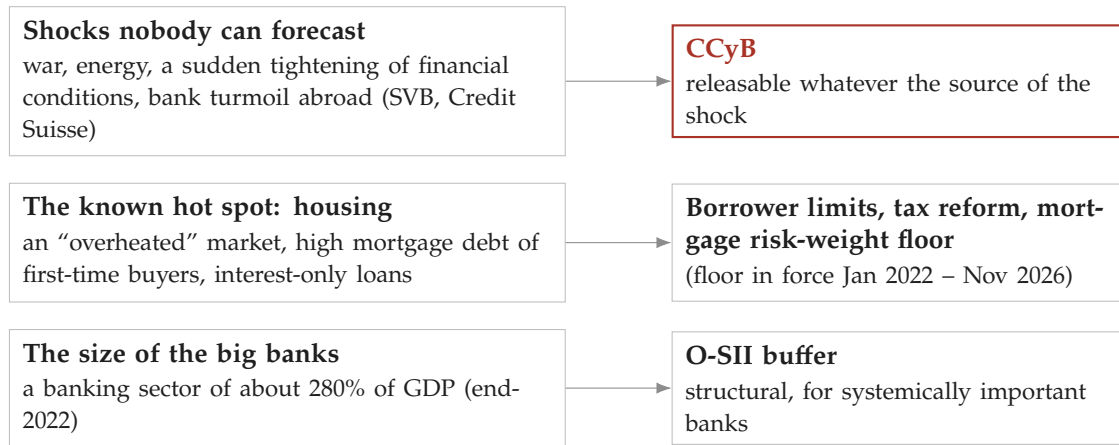
MAY 2023 · FROM 1% TO 2%

*“Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk.”*

DNB cited investors' rising appetite for risk, falling real-estate prices, the debt sustainability of firms and governments, and banks that were “robust, partly due to rising interest rates.”

Both decisions rested on the same three pillars: a normal-to-elevated risk picture, strong banks and the 2020 promise. Credit growth was never the reason.

The buffer was meant for shocks nobody can forecast, while the known hot spots got their own tools



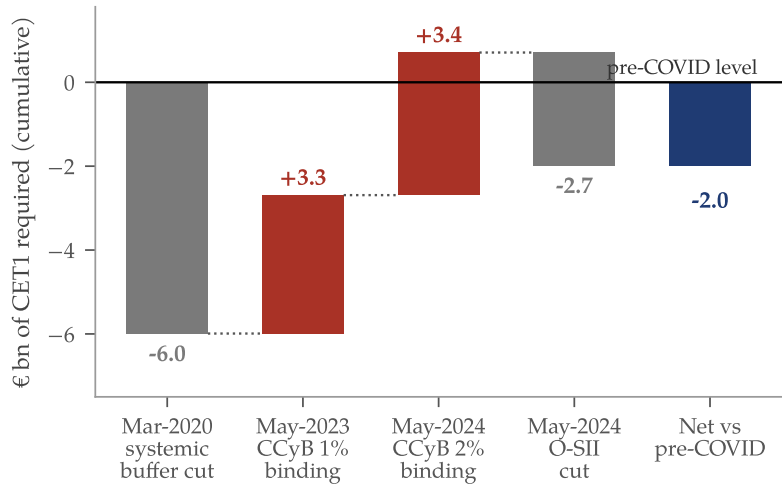
One instrument per target, as the Tinbergen principle suggests. If housing had been the target, the CCyB would be the wrong tool, because it raises the cost of all lending, not only mortgages.

Tested against the evidence, every prediction fits normalisation and none fits overheating

	If DNB was normalising	If DNB was fighting a boom	What actually happened	
Credit indicators	the buffer ignores them	the buffer rises with them	the indicators fell while the buffer rose	✓
The path	+1pp a year, stop at 2%	follows the risk	1%, then 2%, then held	✓
2023: prices falling, rates jumping	raise anyway	pause	DNB raised	✓
2024–26: house prices up 10% again	stay at 2%	go above 2%	held at 2%	✓
Other buffers	cut structural buffers in return	keep them	cut in 2020 and 2024	✓

Five predictions, five matches for **normalisation**. The last row is the key to the next slide.

And the refill was a swap, not a squeeze: capital moved from structural buffers into the releasable one



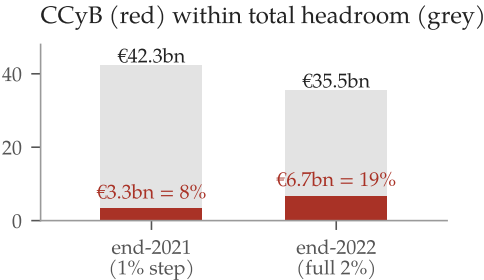
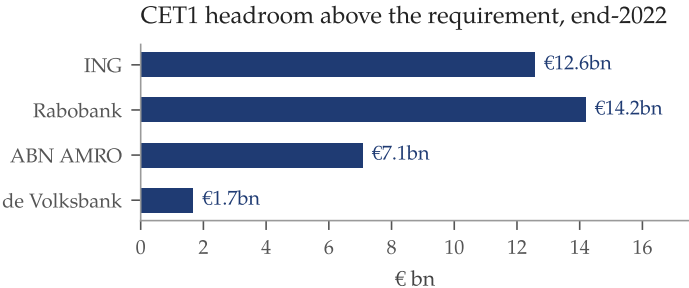
By our estimate, the 2020 cut released about €6bn and the 2024 cut another €2.7bn, while the two CCyB steps added €3.3bn and €3.4bn. Measured against the pre-COVID requirement, banks hold roughly the same amount of capital, or slightly less.

DNB described the swap as “more or less capital-neutral for the three large banks” (2020) and later as “a limited increase in the net capital requirements” (Knot, 2023).

Capital moved from buffers that cannot be released into one that can. Relative to 2021, however, it was a modest increase.

Source: Own calculation (cuts valued at end-2022 RWA of the four largest banks; CCyB amounts: DNB, sector-wide); DNB (17 Mar 2020); Knot (7 Jun 2023).

So was it the wrong moment? No. Higher rates gave banks the profits to build the buffer cheaply

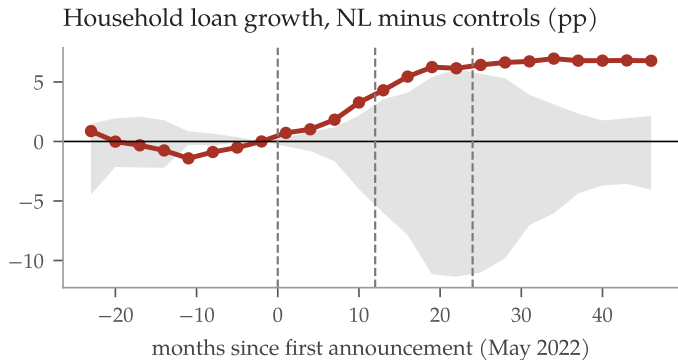
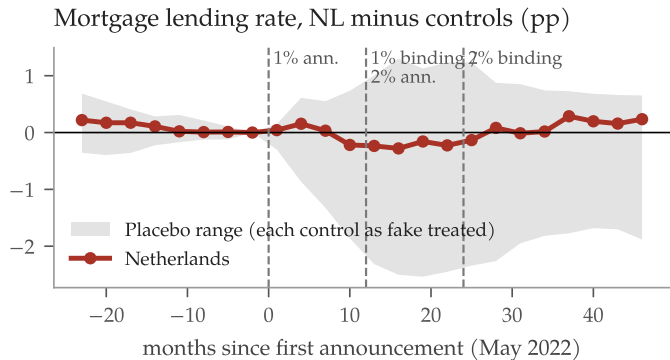


Headroom is the core capital a bank holds above its requirement. The four largest banks had €35.5bn of it at the end of 2022, so the full buffer of €6.7bn used at most a fifth. Higher net interest income meant the buffer could be built from retained earnings rather than by cutting loans, and the ECB finds that a gradual build-up by profitable banks keeps the costs low.

It was not the wrong moment. Arguably it was the right one.

Borrowers did not pay either: credit kept flowing and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



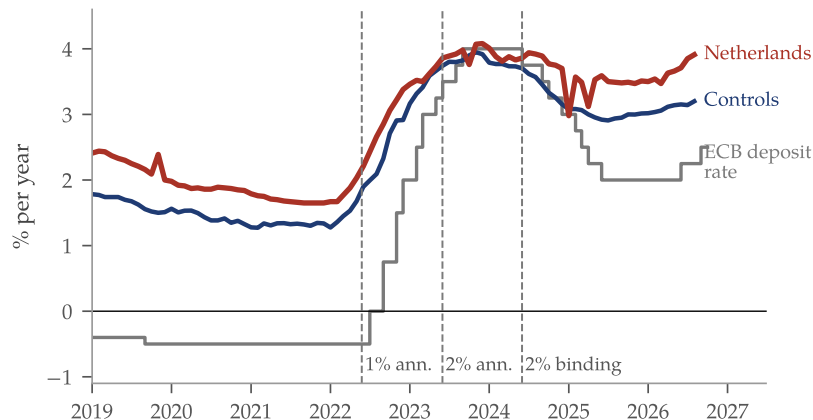
We compare the Netherlands with five euro-area countries that never changed their buffer (Austria, Finland, Italy, Malta, Luxembourg). The red line is the Dutch difference; the grey band shows differences that arise by chance when each control country is treated as if it had raised its buffer. Mortgage rates stayed inside the band, and household credit grew faster than in the controls, partly as a catch-up.

We find **no sign of contraction**, although this is not a causal estimate. DNB reached the same view in September 2026.

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); DNB press release 17 Sep 2026; own two-way fixed-effects event study with placebo inference.

Nor did the buffer double the ECB's squeeze. It let monetary policy focus on inflation

Q3 · IMPLICATIONS



A natural worry is that two tightenings at once, higher policy rates and higher capital requirements, would choke credit. The chart shows that Dutch mortgage rates followed the ECB's rate and moved closely with those in the control countries.

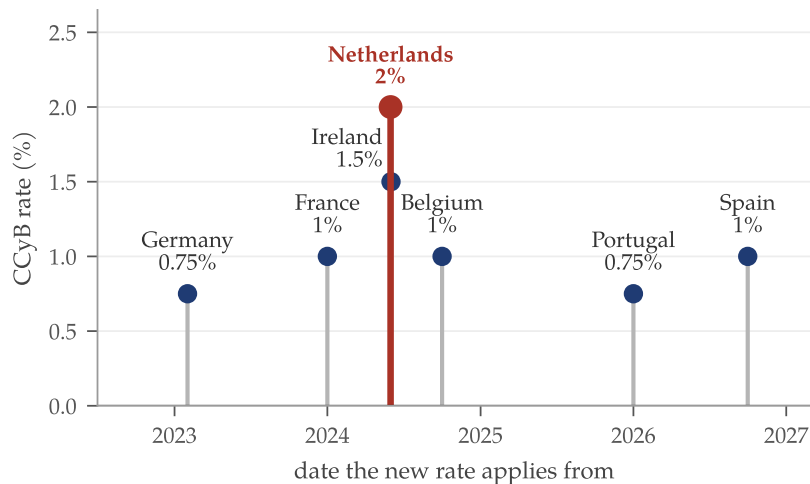
The ECB argues that early CCyB activation *"helps monetary policy focus on its primary objective of price stability."* (ECB, 2025)

With one monetary policy for twenty countries, the CCyB is the Dutch-specific lever, and it is insurance in case a rate shock goes wrong.

Source: ECB Data Portal (MIR mortgage rates; deposit facility rate); ECB Macroprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025). Controls: AT, FI, IT, MT, LU.

Through reciprocity the 2% also binds foreign lenders, and the rest of Europe is moving the same way

Q3 · IMPLICATIONS



Under EU reciprocity, foreign banks that lend into the Netherlands must also hold the Dutch 2% on those loans. This limits leakage to foreign lenders, a known weakness of national capital rules, although lending can still move to non-banks.

The Netherlands was early and set the highest rate, but it was not alone. Since 2023 Germany, France, Ireland, Belgium, Portugal and Spain have all raised their buffers.

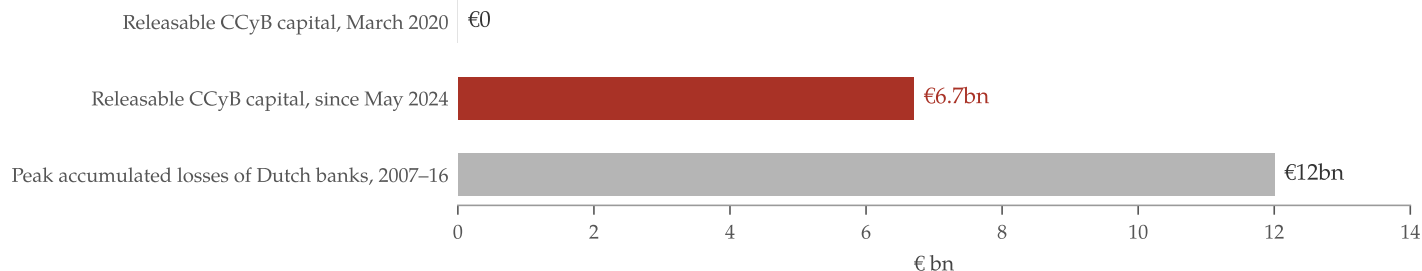
The Dutch move is part of a broader European shift towards keeping releasable capital in normal times.

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macropprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014).

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The result: next time, DNB can release €6.7bn overnight

Q3 · IMPLICATIONS



In March 2020 the Netherlands had no releasable capital. Today it has €6.7bn, about half of the peak losses after the financial crisis and enough to support up to €150bn of lending. Research suggests the payoff is asymmetric: a one-point increase in normal times reduces lending by only about 0.1%, while a one-point release in bad times can raise it by up to 10%.

With the mortgage risk-weight floor expiring in November 2026, DNB says this “*underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.*”

The buffer is cheap to hold and powerful to release.

Source: DNB framework (2022); Lang & Menno (2023), ECB WP 2828; DNB press release 24 Apr 2026.

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Credibility	Without a mechanical rule, markets must trust DNB’s judgement, so DNB has to explain every decision. This is the classic trade-off between rules and discretion (Kydland & Prescott, 1977).
Calibration	Why 2% and not 1.5%? The ECB’s loss-based estimates range from 1.1% to 1.8%, so the Dutch rate sits at the top of the banking union.
Untested	The buffer has never been released. Will banks lend the freed capital, or hoard it, as many did with usable buffers in COVID?
Baseline	Compared with 2021 rather than 2019, the refill <i>was</i> an increase: €3.3bn before the O-SII offset.
The same profits, twice	The ECB lists the Netherlands among the countries that adopted levies on banks’ excess profits in 2022–23. These levies partly claim the earnings that fund the buffer.

None of these overturns the verdict, but they define what DNB must get right next time.

In short, the promise was kept: the buffer was refilled in a storm, at the moment it was cheapest

CONCLUSION

Q1 · ENVIRONMENT

A recovered economy was hit by war, inflation and rate hikes. Credit was cold, housing was hot and then cooled, and the banks were strong and profitable.

Q2 · CRITERIA AND RISKS

A 2% normal level, calibrated on past losses, insures against shocks nobody can forecast. Housing went to other tools, and structural buffers were cut in return.

Q3 · IMPLICATIONS

The refill was cheap and largely a swap, with no sign of contraction. It complements monetary policy, binds foreign lenders and leaves €6.7bn releasable, at the price of discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it, even in a storm.

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake “treated” country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: $(\text{CET1 ratio} - \text{CET1 requirement}/\text{MDA}) \times \text{total RWA per bank}$; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

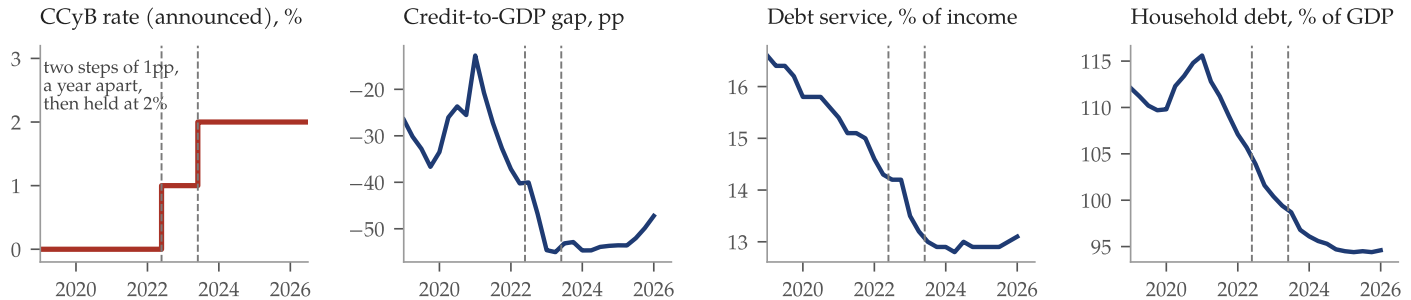
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Appendix A1 · The verdict, rows 1 and 2: the buffer rose in two fixed steps while every credit indicator fell



Row 2 · THE PATH

Normalisation predicts one percentage point a year and a stop at 2%. DNB announced 1% in May 2022 and 2% in May 2023, and has held 2% ever since.

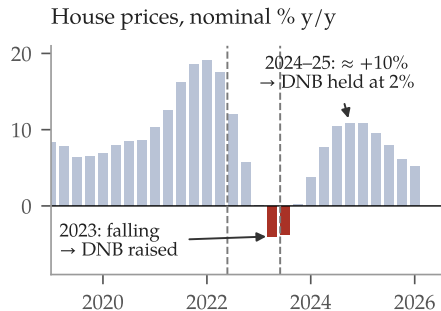
The buffer moved on a timetable, not with credit.

Row 1 · CREDIT INDICATORS

A boom fighter raises the buffer when credit indicators rise. Between the two announcements (dashed lines) the gap, the debt-service ratio and household debt all fell.

Source: ESRB CCyB rates table; DNB press releases (27 May 2022, 31 May 2023); BIS (WS_CREDIT_GAP, WS_DSR, WS_TC). Dashed lines: CCyB announcements.

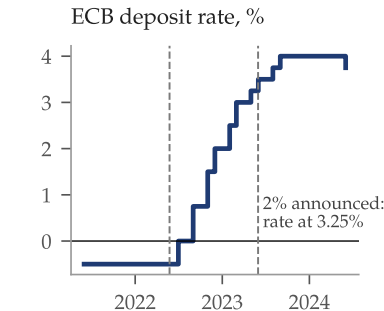
Appendix A2 · The verdict, rows 3 to 5: DNB raised into falling prices, held when prices boomed, and cut other buffers



Row 3 · 2023

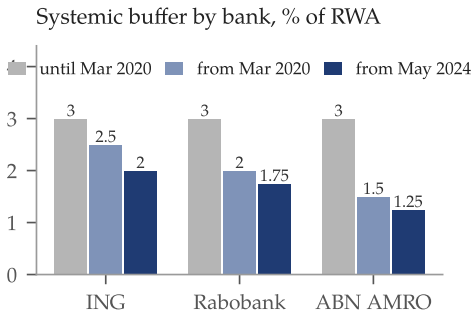
Prices were falling and the ECB rate stood at 3.25%. A boom fighter would pause; DNB raised.

All five rows point the same way: **normalisation, not tightening.**



Row 4 · 2024–26

House price growth returned to about 10% a year. A boom fighter would go above 2%; DNB held at 2%.



Row 5 · OTHER BUFFERS

DNB cut the large banks' systemic buffers in 2020 and again in 2024.